
Evaluate the Current System Landscape

Given an existing system landscape, analyze for constraints and/or pain-points to satisfy a business requirement(s).

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Business Scenario

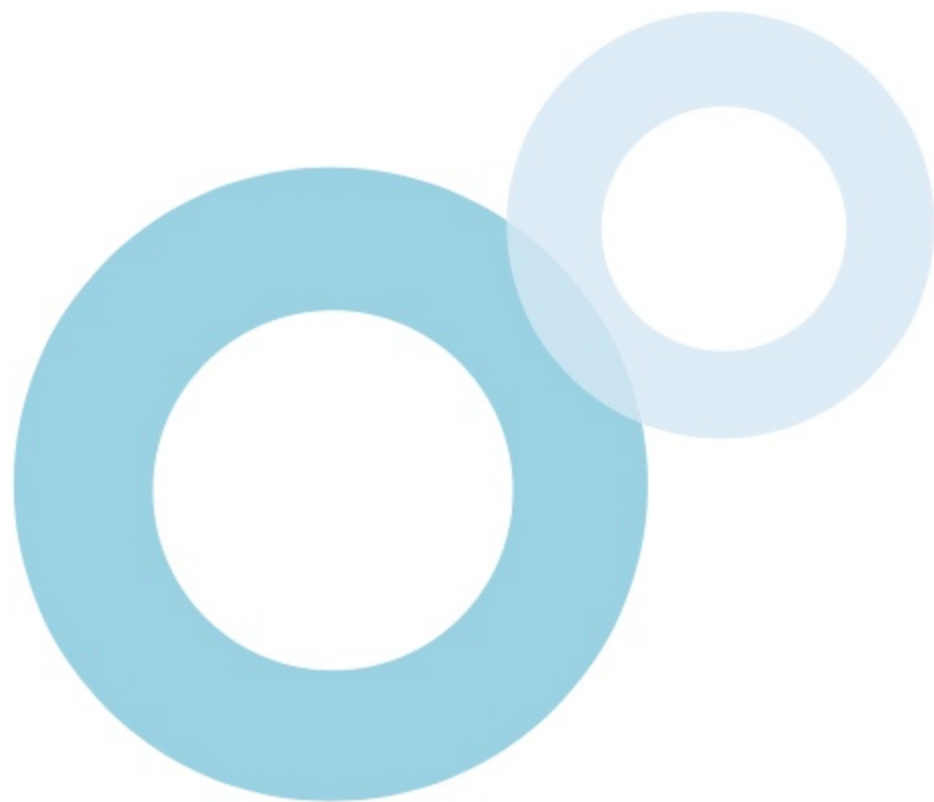


Requirements & Solutions



After studying this topic, you should be able to:

-  Identify how constraints and pain points can affect integration solutions.
-  Analyze for constraints and/or pain points in the given system landscape to satisfy the business requirements.



Introduction

A system landscape may consist of **constraints** and/or **pain points** that can negatively affect integration solutions. For instance, if the **custom logic** defined in an **Apex class** that performs a **SOAP callout** to an ERP system is incorrect, the integration can cause errors in the ERP system.

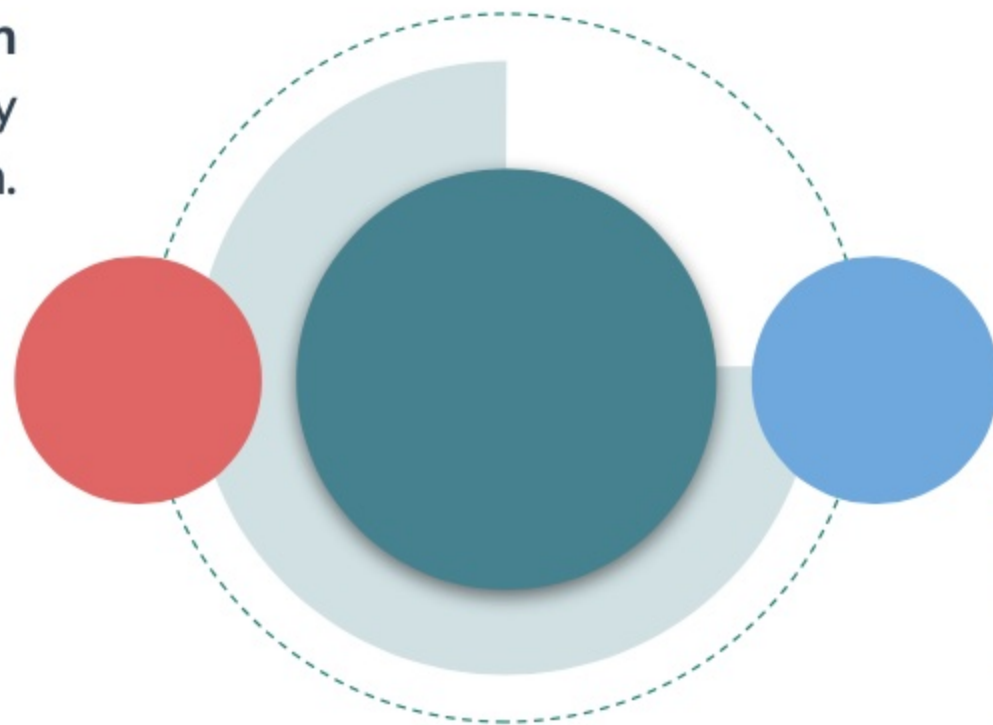
Salesforce features such as **organization-wide defaults**, **profiles**, **permission sets**, and **automated business processes** may need to be assessed to identify any constraints or pain points that can be eliminated.



Overview

CONSTRAINTS

An integration component or feature with a certain **limitation** or **restriction** can negatively affect an integration.



PAIN POINTS

An integration component or feature may have an **issue** or **cause a specific problem** that has an adverse impact on the integration solution.

Constraints & Pain Points

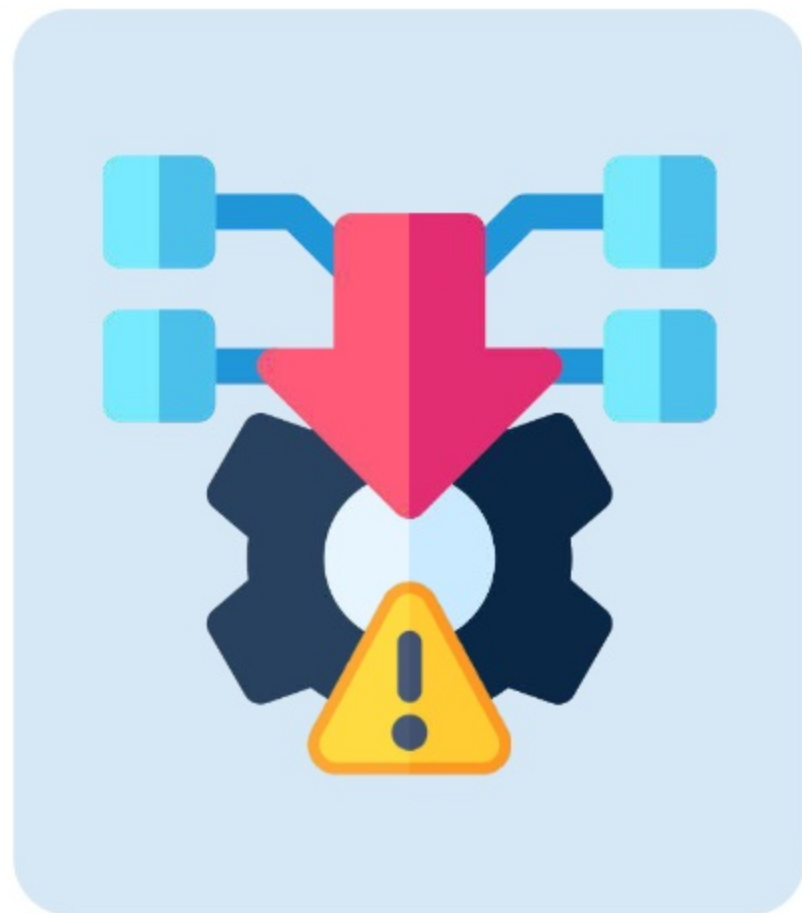
An **integration component or feature** in a system landscape may have a **constraint or represent a pain point** that negatively affects an integration solution.

❖ CONSTRAINTS

A **limitation or restriction** of an integration component or feature can negatively affect an integration. For example, if an integration requires more than **100 synchronous Apex REST callouts** to an external application in a single transaction, the **governor limit for the total number of Apex callouts** in a transaction (**100**) can limit the number of callouts executed.

❖ PAIN POINTS

An integration component or feature may have an **issue or cause a specific problem** that has an adverse impact on the integration solution. For example, if the **custom logic defined in an Apex class** that performs a **SOAP callout to an ERP system** is incorrect, the integration can cause errors in the ERP system.

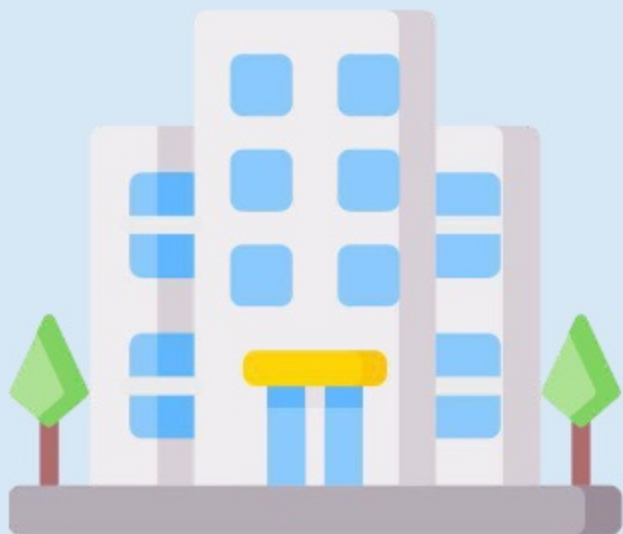


Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Innovation is a company in the United States that sells smart home solutions to thousands of customers every day. Salesforce is mainly used by the company to manage records such as accounts, contacts, and opportunities. It also uses other enterprise systems, including an ERP (Enterprise Resource Planning) system and an SCM (Supply Chain Management) system. Certain external applications are also used, such as for security management. Salesforce has been integrated with the other enterprise systems and external applications. The solution architect is required to analyze the existing integrations for constraints and/or pain points.

Requirements & Solutions

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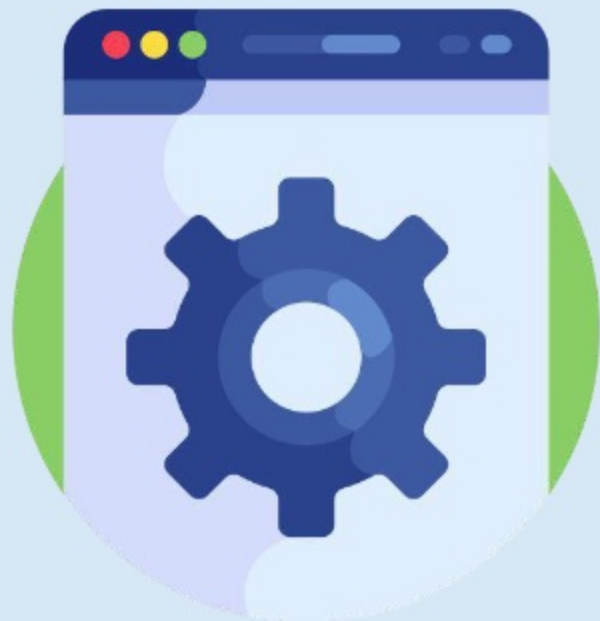


Requirement 1



SCENARIO

Sales reps can access and update records such as accounts, contacts, and opportunities. System administrators have access to an external security management app that allows them to change sharing settings and give specific users 'Read' or 'Read/Write' access to records. However, it has been noticed that the sales reps who have not been given record access can suddenly edit all the opportunity records. All the sales reps have 'Edit' access to the Opportunity object via the same profile. The solution architect needs to recommend a suitable assessment approach that can be used to identify and eliminate the issue.

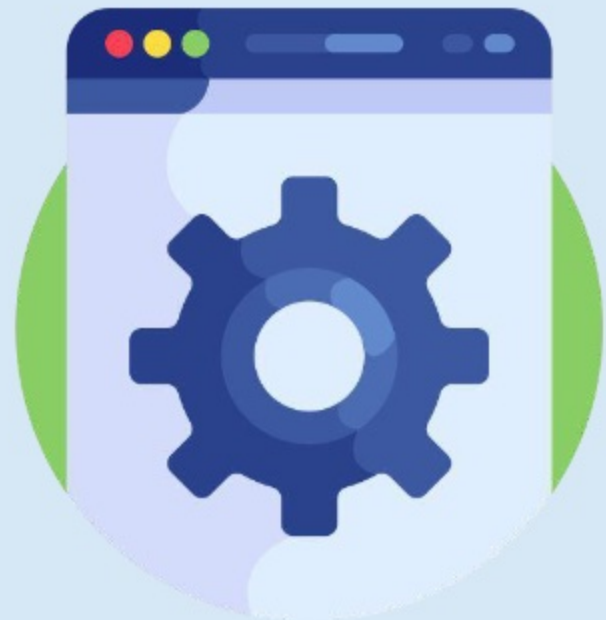


Solution 1



SOLUTION

The **organization-wide default setting** of the **Opportunity** object should be checked to ensure that it is not set to **'Public Read/Write'** since that would give all the sales reps the ability to view and edit not only their own Opportunity records, but also everyone else's opportunities. It is important to also note that the **'Modify All' permission** on an object will allow users to be able to access and edit all records of the object. So, this can be determined by checking the associated **profile** and **permission sets** assigned to the users.

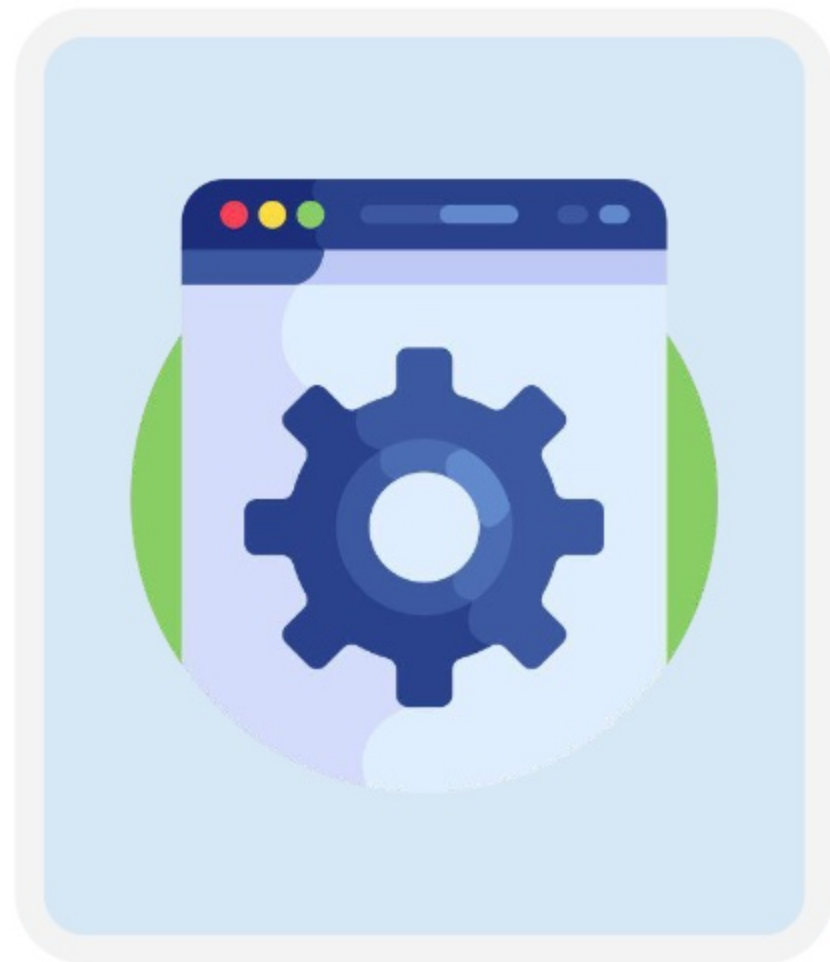


Requirement 2



SCENARIO

The company uses an external application to streamline the security and sharing management of the data stored in multiple enterprise systems, including Salesforce. Certain authorized users can access the external application to view, create, and update profiles, permission sets, and sharing rules via SOAP API. However, the data steward has received complaints that the external application could be causing data security issues in Salesforce. Some associates in the sales department who should not have access can view accounts and contacts. A suitable Salesforce feature must be analyzed to identify and eliminate the issue.

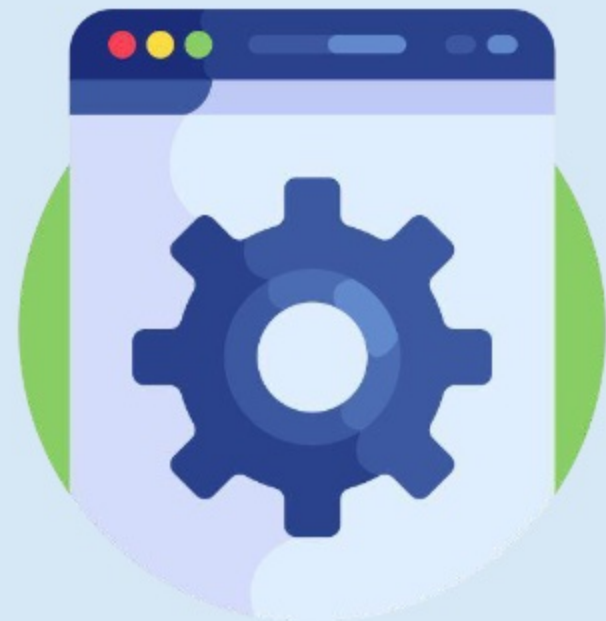


Solution 2



SOLUTION

Profiles and permission sets assigned to the associates should be checked to ensure they do not grant '**Read**' access to the **Account and Contact** objects. **Organization-wide defaults and sharing rules** are only applicable for checking record-level access. In this case, the base-level access to the objects granted via profiles and permission sets needs to be considered.

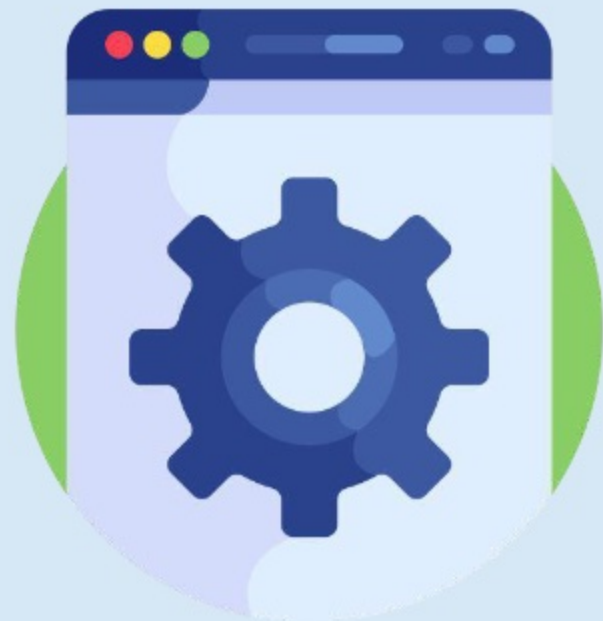


Requirement 3



SCENARIO

A third-party app has been downloaded from the AppExchange marketplace to allow users to integrate Salesforce data such as leads with external data sources. Record-triggered flows have been defined on the Account and Opportunity objects to receive data from the ERP system using External Services and update Salesforce records. Validation rules have been created on several standard and custom objects, including Account, Contact, Opportunity, and Lead. Page layouts and record types have been customized on all the standard objects according to business requirements. However, end users have noticed invalid and incomplete field values on various account and lead records. The solution architect needs to identify the most likely reasons.

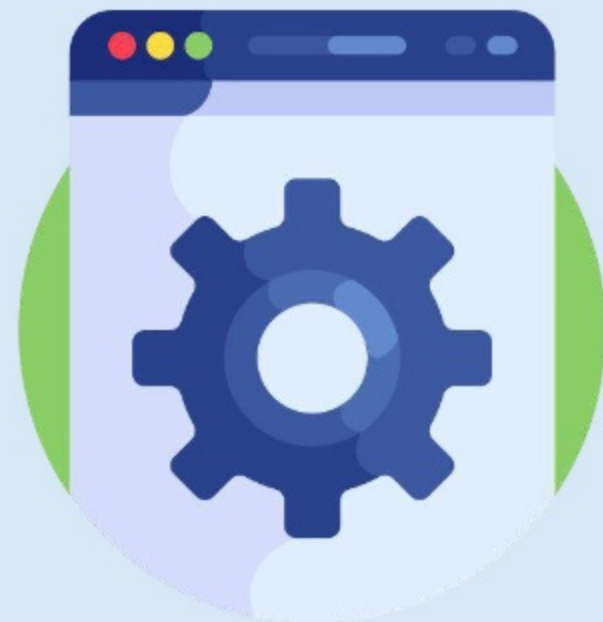


Solution 3



SOLUTION

Given the system landscape and the customizations implemented by the company, the most likely reasons for invalid and incomplete field values would be the **integration via the third-party app** and the **record-triggered flow that receives data from the ERP system**.

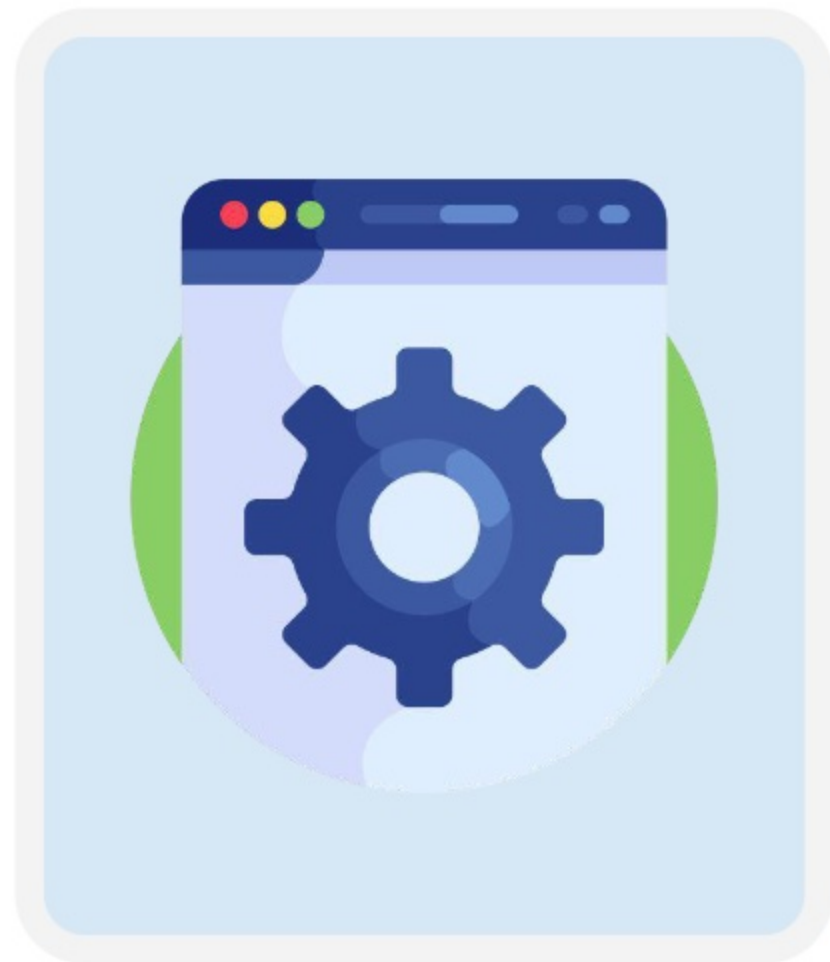


Requirement 4



SCENARIO

Cosmic Grocery uses Salesforce for opportunity management and an ERP system that supports REST API for order management. When an opportunity is closed, asynchronous Apex is used to send the closed opportunity to the ERP system for order creation. When an order is updated in the ERP system, order data is sent to an Apex REST service to update the order information on the related opportunity. A scheduled Apex job checks all closed opportunities daily and sends opportunity information to the ERP system if there are any field updates. A Lightning web component has been configured to receive a platform event message when an order is updated in the ERP system. Sales reps responsible for order management have noticed several orders in the ERP system that should not exist. One of the implemented features must be analyzed to identify and eliminate the issue.

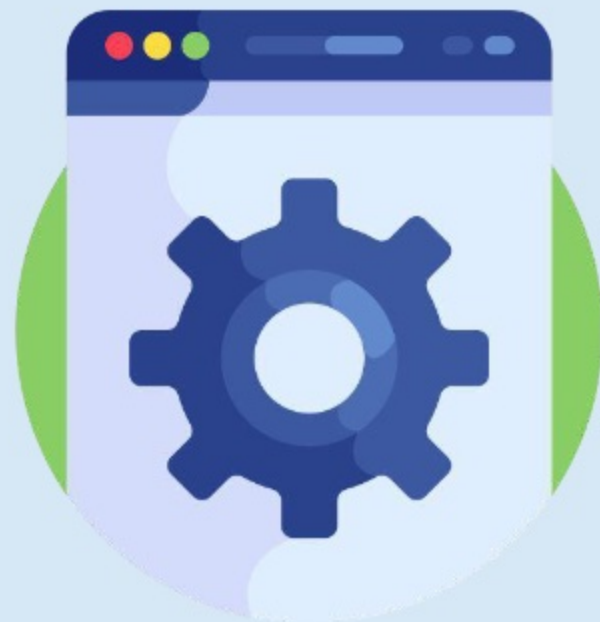


Solution 4



SOLUTION

The **asynchronous Apex** code that sends the closed opportunity to the ERP system should be checked to ensure that it contains the correct logic for order creation. The Apex REST service, scheduled Apex, and Lightning web component do not need to be assessed since they do not participate in order creation.



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